

Profile

M.S. candidate in Chemical Engineering at NTUST, specializing in **electrochemistry**, **ultra-low-temperature electrolytes** and **lithium battery reliability / aging**; co-author on a **Nature Energy** submission.

Research & Industry

Long-Life Li Battery Evaluation for Smart Water Meters

Partner: a smart-water-meter battery maker **Project lead**

Screening methodology

Designed full-stage test protocols, experimental workflows and statistical validation models.

Advanced analysis

EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.

Database & SOP

Established the test database and standardized high-throughput screening procedures (SOP).

Training & tech transfer

Trained QA and R&D teams to independently run large-batch screening and data interpretation.

Nature Energy

Journal submission — battery thermal stability study

Status Under review

Co-author

Responsible for TGA thermal measurement

Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

Experience

- 2025 – Present

Sustainable Electrochemical Energy Development (SEED) Center, NTUST

Lab Equipment Procurement & Asset Maintenance Manager

 - Procurement (>NTD 2M):** Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes and low-temperature baths.
 - Lab safety:** monitor glovebox O₂ / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak.
 - Hazardous materials:** handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li-SOCl₂ cells.
 - TGA testing service:** Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST.
 - Instrument maintenance:** Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.
- Summer 2022

New Taipei City Dept. of Urban Development

Summer Intern, Housing Development Section

 - Reviewed rental subsidy applications and indexed confidential records in government databases; handled frontline public inquiries, building strong familiarity with administrative workflows and SOPs.
- 2021 – 2024

Tunghai University Student Housing / GEPT

Resident Assistant (2 yrs) / Exam Staff (part-time)

 - Oversaw safety and daily order in a 100-resident dormitory, mediating resident disputes and liaising with university administration; as part-time exam staff, handled test-paper check-in with zero errors.

Conferences & Training

- Presenter

2026 TwiChE Annual Meeting

2026/12/05–06 • Feng Chia University

Title: *Weakly Solvating Allyl Ether Electrolytes for Subzero Li || SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability*
- Vendor seminar

TA Instruments ARES-G3 launch & applications forum

2026/07/17 • National Taiwan University
- Vendor lecture

TA Instruments TGA lecture

2026/06/03 • National Taiwan University

Skills & Instruments

Electrochemistry & batteries

- Impedance spectroscopy (EIS)
- Battery aging analysis
- Reliability analysis
- Low-temp electrolytes

Characterization

- Thermal analysis (TGA / DSC)
- Synchrotron (TLS20A / TPS27A / 45A)
- Rheology (ARES-G3)
- Viscometer & densimeter (DMA 5002)

Process & operations

- Lithography (nanoimprint / self-assembly)
- Equipment procurement & maintenance

Software

- Aspen Plus V14
- Aspen Dynamics
- Microsoft Office
- OriginPro
- TA Universal Analysis
- TA TRIOS

Education

- 2025 – 2027 (expected)

National Taiwan University of Science and Technology

M.S. in Chemical Engineering

Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.
- 2020 – 2024

Tunghai University

B.S. in Chemical & Materials Engineering

Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

EHS Licenses

- ✓

Dedicated Wastewater and Sewage Treatment Specialist (Class B)

Certified 2026
- ✓

Indoor Air Quality Maintenance Management Specialist

Certified 2026
- ✓

Waste Disposal Technician (Class A)

Certified 2026
- Occupational Safety and Health Management Personnel (Level B)

2026 / 11 • mid-Nov exam scheduled
- Occupational Safety Management Specialist (Level A)

2027 / 03 • exam scheduled

Instruments

- Maintenance
- TGA Q500
- DSC 2500
- Operation
- ARES-G3
- Anton Paar DMA 5002
- Glovebox
- Procurement
- Glovebox O₂ sensor
- DSC sample press
- Hydraulic electrode punch
- Vacuum pump
- Eppendorf Pipette
- Low-temperature bath